Preliminary Progress Review (PPR)

R26-CS-013

Research Cluster: IAS

NeuraShield:

An Autonomous AI-Driven Cyber Defense Platform with Multi-Layer Threat Detection and Automated Response

BSc (Hons) in Information Technology Specializing in Cyber Security

Supervisor: Professor Harinda Fernando

Co-Supervisor: Ms. Ayesha Wijesooriya

Submitted by:

IT22156556	Deshapriya D.L.N.S.
IT22076670	Mithara P.L.B.
IT22293862	Jayasinghe J.A.D.S.
IT22293312	Uduwanage T.N.

Date: 13th May 2026

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1 Introduction

1.1 Purpose of the Document

The purpose of this Preliminary Progress Review (PPR) document is to present the work completed by the team up to Progress Presentation 1 (PP1) for the final year research project. It provides a formal record of the current status of the project, including literature reviewed, methodology adopted, progress made on each layer, challenges faced, and the plan for the remaining semester. This document is intended for the project supervisor, project coordinator, and examination panel to evaluate the project's direction, feasibility, and progress.

1.2 Definitions, Acronyms, and Abbreviations

Acronym	**Full Form**
PP1	Progress Presentation 1
PPR	Preliminary Progress Review
ZTA	Zero-Trust Authentication
BB-MAS	Boston University Multi-Device Multimodal Authentication and Signature Dataset
CNN	Convolutional Neural Network
JSON	JavaScript Object Notation
API	Application Programming Interface
NIDS	Network Intrusion Detection System

1.3 Overview

The remainder of this PPR is organized as follows: Section 2 presents the literature review. Section 3 identifies the research problem and its significance. Sections 4 and 5 state the research questions and objectives. Section 6 describes the research methodology. Section 7 details the data sources and analysis methods. Sections 8-11 cover anticipated benefits, scope, constraints, and project schedule respectively. References and appendices are provided at the end.

1.4 Background and context

The rapid evolution of cyber threats has made traditional perimeter-based security insufficient for modern networks. Attackers increasingly rely on credential theft and session hijacking techniques to bypass initial authentication controls and gain persistent access to systems. Once inside, they can perform lateral movement, deploy ransomware, exfiltrate sensitive data, or maintain long-term presence with minimal detection. This problem is particularly acute in educational institutions and small-to-medium enterprises that often lack dedicated security operations centers and rely on shared infrastructure.

Credential compromise remains one of the most common attack vectors. Phishing campaigns, infostealer malware, and credential-stuffing attacks allow adversaries to obtain valid login credentials, after which they can operate as legitimate users. Even strong multi-factor authentication can be bypassed through real-time phishing or session token theft. As a result, many breaches go undetected for weeks or months, leading to significant damage.

The impact of such attacks is severe and far-reaching. Organizations face financial losses, intellectual property theft, regulatory penalties, and reputational damage. Students and staff in academic environments risk exposure to personal and research data, while small businesses often lack the resources to recover quickly from incidents. According to industry reports, breaches involving lateral movement after credential compromise result in substantially higher costs due to extended dwell time and broader system compromise.

Existing security solutions have notable limitations. Static multi-factor authentication and traditional zero-trust implementations primarily focus on initial access control. Once a session is established, continuous verification of user identity is rarely enforced. Behavioral biometrics tools exist but are typically limited to login-time or transaction monitoring and lack integration-

with real-time threat signals from network, endpoint, or content layers. This creates a dangerous gap where compromised sessions can remain active despite ongoing threats.

NeuraShield was conceived to address these challenges by building a fully autonomous, on-premises cyber defense platform. The system integrates four specialized layers that work collaboratively: network threat detection, endpoint ransomware protection, multimodal content threat detection, and continuous zero-trust behavioral authentication. By combining real-time threat intelligence with adaptive behavioral monitoring, the platform aims to provide proactive, layered defense while maintaining usability and strict data privacy.

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2 Literature Review

2.1 Behavioral Biometrics and Continuous Authentication

Continuous authentication extends traditional verification by passively monitoring user behavior throughout an active session, thereby providing protection against session hijacking and credential theft. The zero-trust security model rejects the assumption of implicit trust after initial login and instead demands continuous validation. Keystroke dynamics, which analyze typing rhythm, dwell time, and flight time patterns, have proven to be one of the most practical behavioral biometric signals for this purpose. Alotaibi and Alghamdi (2023) demonstrated that deep learning models applied to keystroke dynamics can effectively distinguish legitimate users from impostors with high accuracy, supporting its selection as the primary behavioral signal in ZTA Layer of NeuraShield [6].

Multimodal behavioral fusion further enhances continuous authentication performance. Shen et al. (2022) showed that combining keystroke dynamics with mouse dynamics significantly reduces equal error rates compared to using either modality alone, highlighting the complementary strength of multiple behavioral signals [7]. Wang et al. (2023) identified adaptive thresholding and context-aware risk scoring as major open challenges in behavioral biometrics. Their survey emphasized that static thresholds often fail to accommodate natural intra-user variations caused by stress, fatigue, or device changes [8]. This insight directly motivated the implementation of a context-aware adaptive risk thresholding mechanism in Layer A, which dynamically adjusts anomaly detection sensitivity based on user activity context and threat intelligence received from other NeuraShield layers.

2.2 Deepfake and Multimodal Threat Detection

The increasing sophistication of AI-generated threats has made multimodal detection essential for modern cybersecurity. Basit et al. (2021) conducted a comprehensive survey on AI-enabled phishing attack detection techniques and highlighted the effectiveness of machine learning approaches in identifying deceptive emails by analyzing various textual and structural features [2]. Earlier work by Hamid and Abawajy (2011) demonstrated that hybrid feature selection methods can significantly improve phishing email detection accuracy by selecting the most relevant characteristics from email content [1]. These studies underscore the importance of content-based analysis in detecting social engineering attacks that often accompany deepfake videos.

Multimodal threat detection extends beyond single attack vectors by integrating insights from different security domains. Ring et al. (2020) showed that machine learning techniques combined with data visualization can effectively detect ransomware attacks by identifying anomalous behavioral patterns [3]. This behavioral analysis approach, when combined with content-based detection methods for phishing and deepfakes, creates a more robust defense mechanism. The literature supports the development of this Layer in NeuraShield, which adopts a multimodal strategy by fusing textual phishing detection with visual deepfake analysis to provide comprehensive threat coverage.

2.3 Ransomware and Endpoint Protection

Ransomware has become one of the most destructive cyber threats, causing significant financial and operational damage to organizations worldwide. Modern ransomware detection has shifted from traditional signature-based methods to behavioral analysis approaches that monitor file system activities, encryption patterns, and anomalous process behaviors. Research emphasizes the importance of early detection through real-time monitoring of file access rates, entropy changes, and I/O operations, which allow systems to identify ransomware activity before widespread encryption occurs. Layer C (Ransomware Killer) builds upon these principles by implementing behavioral monitoring and automated response mechanisms to detect and contain ransomware attacks at the endpoint level.

2.4 Network Intrusion Detection

Network security remains a critical layer in defending against lateral movement and external attacks. Traditional rule-based intrusion detection systems have limited capability against zero-day and sophisticated attacks. Recent advancements in leverage machine learning techniques to analyze network traffic flows, packet patterns, and communication behaviors to identify anomalies in real time. These approaches enable the detection of unusual data exfiltration, command-and-control communication, and scanning activities. Layer D (Network Guardian) adopts this data-driven methodology to monitor network traffic and generate timely alerts that can be shared with other layers for coordinated defense.

2.5 Identified Gap:

Most existing systems operate in silos and lack real-time cross-layer intelligence and adaptive response mechanisms. NeuraShield addresses this by integrating behavioral authentication, content analysis, endpoint protection, and network monitoring into a unified autonomous platform.

3 Identification and Significance of the Problem

sophisticated and multi-vector attacks. Once attackers obtain valid user credentials, which occur in a majority of breaches, traditional security mechanisms such as firewalls, Multi-Factor Authentication (MFA), and endpoint protection solutions often fail to prevent lateral movement, data exfiltration, ransomware deployment, and social engineering attacks. Current systems typically operate in isolation, lack continuous verification of user identity, and cannot effectively correlate threats across network, endpoint, content, and behavioral domains in real time.

This results in prolonged attacker dwell time, delayed detection, and limited automated response capabilities. Additionally, the rapid rise of AI-powered threats such as deep-fake videos and advanced phishing emails has further complicated detection efforts. There is a pressing need for an integrated, autonomous, and intelligent cyber defense platform that can continuously monitor multiple attack surfaces, correlate information across layers, and respond adaptively in a coordinated manner while maintaining full on-premises operation for data privacy and compliance.

4 Research Questions

1. How can multiple security layers (behavioral, content, endpoint, and network) be integrated to enable real-time coordinated threat detection and response?

2. How can context-aware adaptive mechanisms improve continuous authentication in post-compromise scenarios?

3. What is effective architecture for a fully on-premises autonomous multi-layer cyber defense system?

5 Research Objectives

Main Objective:

To design, develop, and evaluate NeuraShield, an autonomous, multi-layer, on-premises AI-driven cyber defense platform with intelligent threat detection and automated response capabilities.

Specific Objectives:

- Develop a Continuous Zero-Trust Authentication layer using behavioral biometrics with adaptive risk thresholding.
- Implement a Multimodal Content Threat Detection layer for phishing emails and deepfakes.
- Build Ransomware Killer and Network Guardian layers with alert generation.
- Design a cross-layer alert fusion and coordination mechanism.
- Evaluate the system's effectiveness through prototypes and simulated scenarios.

6 Research Methodology

The project follows the Design Science Research (DSR) methodology. This involves problem identification, artifact design and development, demonstration, evaluation, and communication.

Each layer is developed as an independent microservice and later integrated. Agile-style sprints with weekly team meetings are followed. Models are trained using standard benchmark datasets and evaluated using accuracy, precision, recall, and confusion matrices.

7 Sources for Test Data and Analysis

- **Layer A - Continuous Zero-Trust Authentication Layer**
  - **Datasets Used:** BB-MAS Dataset (Boston University Multi-Device Multimodal Authentication and Signature Dataset) keystroke data from 117 users containing over 606,000 keystroke records. Features extracted include dwell time (how long a key is held), flight time (time between key releases and presses), key overlap, typing speed, and session break patterns.
- **Layer B - Multimodal Content Threat Detection Layer**
  - Datasets Used:
    - Text: Nazario, Nigerian Fraud, CEAS_08, SpamAssasin, Enron, Ling (7 benchmark datasets)
    - Video: FaceForensics++ DFD subset (300 real + 300 fake videos)
    - Audio: ASVspoof 2019 Deepfake track
- **Layer C - Endpoint Behavioral Protection Layer**
  - **Datasets Used:** Based on the CIC-MalMem-2022 data set for detecting ransomware attacks, managing the detected ransomware attacks by terminating the malware process, decrypting the files with the help of the pre-saved snapshots, and issuing alerts in JSON format to other layers of NeuraShield.
- **Layer D - Zero-Day Intrusion Detection Layer**
  - **Dataset Used:** Network traffic datasets CIC-IDS2017

7.1 Data Analysis Methods

The project employs a combination of data preprocessing, feature engineering, machine learning modeling, and real-time analysis techniques suited to each layer:

- Feature Engineering: Raw data from different sources (keystroke logs, network packets, video frames, email text, etc.) is processed to extract meaningful features. For example, in Layer A, keystroke dwell time, flight time, key overlap, and session patterns are derived from raw press/release events. In Layer B, video frames are converted into color statistics and edge ratios, while text is transformed using TF-IDF vectorization.

- Anomaly Detection: Used primarily in Layer A (Zero-Trust Authentication) and Layer C (Ransomware Killer). Unsupervised algorithms such as Isolation Forest and One-Class SVM are applied to identify deviations from normal user or system behavior without requiring labeled attack data.
- Supervised Classification: Applied in Layer B for deepfake video detection (CNN models) and phishing email classification (Random Forest with TF-IDF). These models are trained on labeled datasets to distinguish between benign and malicious samples.
- Real-time Alert Correlation: A custom fusion and correlation mechanism is implemented across layers. JSON alerts generated by one layer are consumed by other layers (especially Layer A) to dynamically adjust risk scores and decision thresholds in real time.

7.2 Model Validation Techniques:

- Train-test split (typically 70:30 or 80:20 ratio)
- K-fold cross-validation to ensure model robustness
- Performance metrics including Accuracy, Precision, Recall, F1-Score, and Confusion Matrix
- Real-world simulation testing using dummy attack scenarios to evaluate end-to-end system behavior

This hybrid approach (unsupervised + supervised + real-time correlation) allows the system to handle both known and unknown threats effectively while maintaining adaptability.

8 Anticipated Benefits

8.1 Benefits to Users

- **Organizations and Enterprises**: NeuraShield will provide stronger protection against credential theft and post-compromise attacks. It enables early detection of ransomware, deepfakes, phishing, and network intrusions while reducing false alarms through cross-layer intelligence. This will help organizations minimize financial losses, data breaches, and downtime.
- **Individual Users (Students, Staff, and Employees)**: The system offers continuous protection even after passwords or credentials are stolen. Users will benefit from seamless yet secure authentication that adapts intelligently without unnecessary interruptions, significantly improving both security and usability.

8.2 Contribution to the Body of Knowledge

This project contributes to the field of cybersecurity by demonstrating a practical, fully on-premises autonomous multi-layer cyber defense architecture. It addresses the current research gap in cross-layer integration by combining behavioral biometrics, multimodal content analysis, endpoint protection, and network monitoring into a coordinated, adaptive system.

The findings from this project will be useful for future researchers working on zero-trust architectures, autonomous SOCs, and integrated AI-driven defense systems.

9 Scope and Specified Deliverables / Expected Research Outcome

- **In Scope**: Design and development of four integrated layers, cross-layer alert system, working prototypes with dashboards, and performance evaluation.
- **Out of Scope**: mobile client support, and large-scale production testing.
- **Expected Outcome**: A fully functional prototype of NeuraShield with demonstrated cross-layer coordination.

10 Research Constraints

- Limited hardware resources for model training.
- Time constrains for full integration before PP1.
- Availability of high-quality labeled datasets for all modalities.
- Must maintain fully on-premises deployment (no cloud services).

11 Project Plan or Schedule

- Phase 1 (Completed - PP1): Literature review, individual layer development, basic integration.
- Phase 2 (May - July 2026): Full system integration, central orchestration, real-time hooks, end-to-end testing, final dashboard.
- Final phase: Final optimization, Full system testing.

12 Current Results Summary

All four layers have been successfully implemented as standalone modules. Feature extraction, model training, and basic integration interfaces are complete. The platform demonstrates promising standalone performance and is ready for full system integration in the next phase.

Layer	Component	Major Achievements
A	Zero-Trust Auth	BB-MAS dataset processed, user profiles created, Isolation Forest + One-Class SVM trained, RiskEngine with adaptive thresholding, Fast API + Dashboard
B	Multimodal Detection	CNN model for deepfake video (91% acc on 600 videos), Text phishing classifier (80%+), Audio model, Multimodal fusion engine, Web demo.
C	Ransomware Killer	Implemented real-time detection, automatic process termination, and alerting within six seconds.
D	Network Guardian	Network anomaly detection prototype

13 References

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14 Appendices

Appendix A: System Architecture Diagram

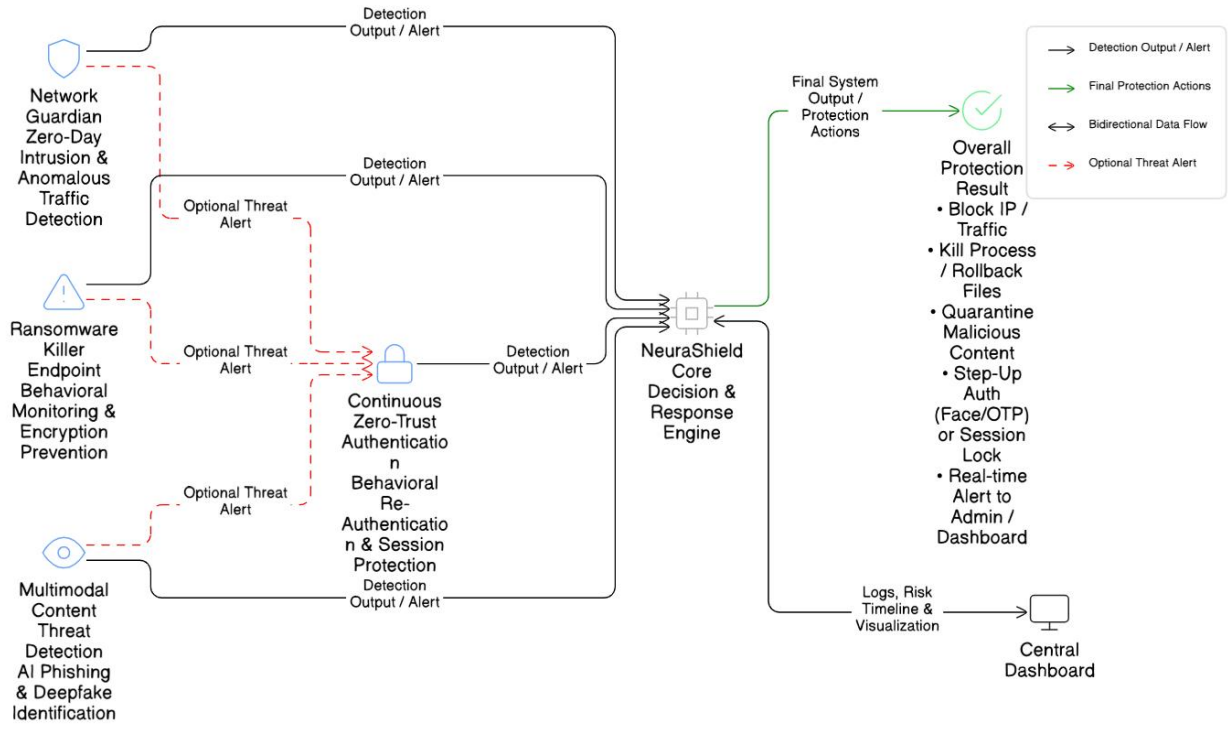


Figure 1: System Architecture Diagram